EST II — MATH LEVEL 2 • UNIT 6.01

C Permutations, Combinations & Conditional Probability

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Permutations, Combinations & Conditional Probability Practice Exercises

1 MCQ

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7+4i}{2-i}$?

A $2 + 3i$

B $2 - 3i$

C $\frac{10+15i}{3}$

D $\frac{18+15i}{5}$

2 MCQ

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?

A 0

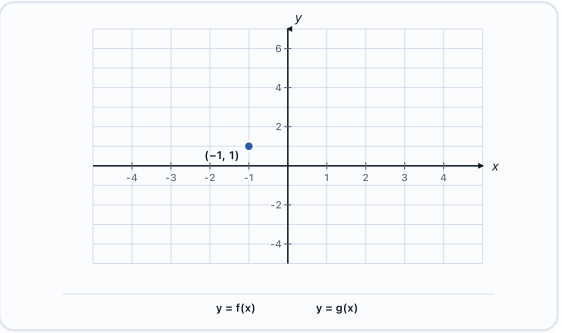
B 2

C -1

D 3

3 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?



A (1, 0) and (3, 0)

B (0, 1) and (0, 3)

C (-1, 0) and (-3, 0)

D (1, 0) and (-3, 0)

4 MCQ

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

A [3, 4]

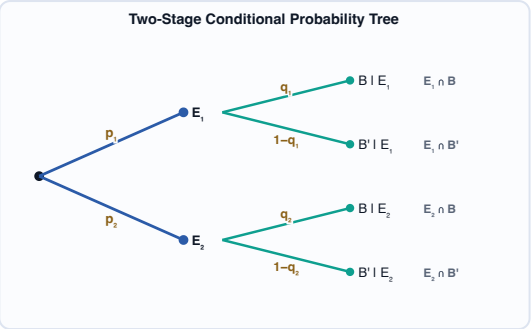
B [5, 8]

C [-3, -4]

D [4, 3]

5 MCQ

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?



A $\frac{3}{5}$

B $\frac{2}{5}$

C $\frac{1}{2}$

D $\frac{3}{4}$

6 MCQ

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

A 14

B 17

C 8

D 11

7 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

A 96

B 192

C 98

D -96

8 MCQ

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.

A 0

B -2

C 2

D 1

9 MCQ

A survey was conducted on a group of 100 students regarding their preferences for two subjects, Mathematics (M) and Science (S). The results are summarized in the two-way table below, where some values are unknown. If a student is chosen at random from this group, the probability that the student prefers Mathematics but not Science, $P(M \cap \neg S)$, is equal to the probability that the student prefers Science but not Mathematics, $P(\neg M \cap S)$. What is the value of x?

	Science (S)	Not Science (¬S)	Total
Mathematics (M)	25	x	45
Not Math (¬M)	y	35	55
Total	45	z	100

A 30

B 15

C 20

D 25

10 MCQ

A manufacturer produces light bulbs. The probability that a randomly selected light bulb is defective is p. The production process is designed such that the probability of producing exactly k defective bulbs in a batch of $n = 5$ bulbs follows the binomial probability formula: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$. It is known that the probability of finding exactly 2 defective bulbs in a batch of 5 is $\frac{10}{32}$. Find the value of p.

A 0.8

B 0.5

C 0.75

D 0.2

11 MCQ

A bag contains 5 red marbles and 3 blue marbles. If a marble is drawn at random, what is the probability that it is blue?

A $\frac{5}{8}$

B $\frac{3}{5}$

C $\frac{5}{3}$

D $\frac{3}{8}$

12 MCQ

A factory produces two types of components, A and B. Components of type A have a defect rate of 3%, and components of type B have a defect rate of 5%. A batch of 100 components is produced, consisting of N components of type A and $100 - N$ components of type B. Let X be the random variable representing the total number of defective components in the batch. If the expected number of defective components is $E[X] = 4.2$, find the probability that a randomly selected component from this batch is defective, given that it is of type A.

A 0.036

B 0.042

C 0.030

D 0.050

13 MCQ

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

A 1/3

B 1/6

C 1/2

D 2/3

14 MCQ

A fair six-sided die is rolled. Find $P(\text{rolling a } 3)$.

A $\frac{1}{6}$

B $\frac{5}{6}$

C $\frac{1}{3}$

D $\frac{1}{2}$

15 MCQ

A fair six-sided die is rolled. Find $P(\text{rolling a } 4)$.

A $\frac{1}{3}$

B $\frac{1}{6}$

C $\frac{1}{2}$

D $\frac{5}{6}$

16 MCQ

In an applied problem, a fair six-sided die is rolled. Find $P(\text{rolling a } 2)$.

A $\frac{5}{6}$

B $\frac{1}{3}$

C $\frac{1}{6}$

D $\frac{1}{2}$

17 MCQ

Suppose a fair six-sided die is rolled. Find $P(\text{rolling a } 1)$.

A $\frac{1}{3}$

B $\frac{1}{6}$

C $\frac{1}{2}$

D $\frac{5}{6}$

18 MCQ

An observed fair six-sided die is rolled. Find $P(\text{rolling a } 5)$.

A $\frac{5}{6}$

B $\frac{1}{3}$

C $\frac{1}{2}$

D $\frac{1}{6}$

19 MCQ

Consider a fair six-sided die is rolled. Find $P(\text{rolling a } 6)$.

A $\frac{1}{3}$

B $\frac{1}{6}$

C $\frac{1}{2}$

D $\frac{5}{6}$

20 MCQ

A fair six-sided die is rolled. Find $P(\text{rolling a number less than } 3)$.

A $\frac{1}{3}$

B $\frac{1}{2}$

C $\frac{2}{3}$

D $\frac{1}{6}$

21 MCQ

A fair six-sided die is rolled. Find $P(\text{rolling a number greater than } 4)$.

A $\frac{1}{2}$

B $\frac{1}{3}$

C $\frac{2}{3}$

D $\frac{1}{6}$

22 MCQ

If $P(E) = \frac{14}{19}$, find $P(E')$.

A $\frac{5}{19}$

B $\frac{14}{19}$

C $\frac{6}{19}$

D 1

23 MCQ

Suppose that $P(E) = \frac{3}{11}$, find $P(E')$.

A $\frac{8}{11}$

B $\frac{3}{11}$

C $\frac{9}{11}$

D 1

24 MCQ

Assuming that $P(E) = \frac{4}{19}$, find $P(E')$.

A $\frac{15}{19}$

B $\frac{16}{19}$

C 1

D $\frac{4}{19}$

25 MCQ

Under the condition that $P(E) = \frac{8}{9}$, find $P(E')$.

A $\frac{8}{9}$

B $\frac{2}{9}$

C 1

D $\frac{1}{9}$